Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/05/2018
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Colin, Catherine Ph.D. (Institut de Mécanique des Fluides de Toulouse)		
Grant/Contract No.:	NNX15AK16G		
Performance Goal No.:			
Performance Goal Text:			

EDITOR'S NOTE (5/17/2016): Continuation of "Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry" with same Principal Investigator, Dr. Jungho Kim, grant # NNX12AF31A. Added to Task Book when received information.
This effort will be conducted with the European Space Agency (ESA) in one of the facilities/test articles that they are developing.

Fluorescent paints that are sensitive to oxygen concentration (pressure sensitive paints, PSP) and temperature (TSP) have been used to measure shear stress, surface pressure, and wall temperature distributions in aerodynamic and fluid mechanics applications since the 1980s. It has seen very limited use for phase-change applications, however. If only temperature is to be measured, the paint is encapsulated within a binder or between impermeable barriers to eliminate sensitivity to oxygen. The molecules are typically stimulated using blue-UV light, and emission at a longer wavelength

Task Description:	occurs. The intensity of the emitted light decreases with temperature and oxygen concentration, and a filter is used to remove the excitation light before the emitted light is measured by a camera. The emission lifetime is typically on the order of microseconds, much faster than the boiling processes being measured, so the system frequency response is usually limited by the thermal response of the materials onto which the paint is bonded. The advantages of TSPs over infrared thermography and other techniques such as microheater arrays or microbolometers to measure temperature are numerous. Readily available CCD or CMOS cameras can be used to measure the intensity distribution instead of expensive IR cameras. Since TSPs respond very quickly, the transient response of TSP based measurements is limited in practice only by the speed of the camera, which are typically much faster than IR cameras. Also, since visible light is used for illumination and sensing, glass, clear plastics, PDMS and other commonly used materials can be used in place of expensive and difficult to machine IR transparent materials (e.g., silicon, sapphire, CaF2) as substrates. Calibration of TSPs can be performed without having to consider emission from the transparent substrate they are placed on, unlike for IR measurements. Finally, the wavelength of the light used (~0.5 mm for TSPs) is much smaller than the 3-5 µm or 8-12 µm light IR cameras are sensitive to, allowing much higher spatial resolution measurements. The diffraction limit for 0.6 µm light in air is ~0.3 µm. TSPs have quite high sensitivity (approximately -2%/°C), and TSPs are available that can be used over a wide temperature range. Although TSPs have been used for aerospace applications for many years, very little work has been performed using this technique for two-phase flows. We have developed methods whereby temperature and heat flux distributions can be measured during pool boiling and flow boiling so models of pool and flow boiling can be verified. A test apparatus has been developed and this technique will be used to measure gravity effects on flow boiling during a series of low-g aircraft flights in March, 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	Gravity can have a significant effect on flow boiling heat transfer performance. Past research using parabolic aircraft has resulted in better understanding of how flow boiling is altered in microgravity for a variety of test section geometries, working fluids, and flow regimes. The goal of this research is to be able to predict the effect of changing gravity levels on heat transfer as well as conditions which are independent on gravity level by studying the local effects of different flow boiling conditions at low mass flux and low heat flux where differences between gravity levels have been observed in the past. Local measurement techniques, which allow measurement of the wall heat transfer distribution, are desired so heat transfer mechanisms can be identified. There is a strong need for a low-cost, optical measurement technique by which local heat transfer coefficients can be measured. Fluorescent paints that are sensitive to oxygen concentration (pressure sensitive paints, PSP) and temperature (TSP) have been used to measure shear stress, surface pressure, and wall temperature distributions in aerodynamic applications since the 1980s (e.g., Campbell et al. (1994), and Ozawa et al. (2015)). To measure temperature alone, the sensitivity to oxygen is eliminated by encapsulation or removal of oxygen from the system. The paints are typically excited by blue/UV light and fluoresce at a longer wavelength. As temperature increases, the intensity of the emitted light decreases and the peak in the spectrum shifts to longer wavelengths. TSP calibration can be performed in-situ which eliminates intensity variation due to concentration. In this work, temperature sensitive paints (TSP) are used to measure the local heat transfer distribution during flow boiling of HFE-7000 in a 6 mm ID x 120 mm long tube during upflow, downflow, with laminar inlet flow conditions and low heating rates at various gravity levels. Heat transfer distribution and flow visualization were performed simultaneously using two high-speed CCD cameras. Data were obtained at four mass fluxes, five heat fluxes, and nine gravity levels, and observed differences in wall heat transfer trends were investigated. A paper describing the technique was published in the Journal of Heat Transfer in October, 2017 (printed publication in March 2018--see Bibliography section). The work performed between May, 2017 through April, 2018 was concerned with reducing microgravity flow boiling data obtained during two days of flying in the Zero-G aircraft in March, 2017, taking additional data in November, 2017 during two days of flying in November, 2017, and reducing/analyzing this data. The test apparatus was modified over the summer of 2017 so data could be obtained in upward as well as downward flow by simply turning a 4-way valve. The November 2017 flights were used to obtain data in high-g, Martian, and Lunar gravity in upward and downward flow directions along with microgravity data. The effects of wall heat flux, mass flux, and subcooling were investigated. A paper summarizing the results is currently being written and will be submitted to the Int. J. of Heat and Mass Transfer. This data indicates that a reduction of gravity causes a deterioration in the heat transfer compared with upward/downward flow due to the reduction in slip velocity between the bubbles and the bulk fluid. The gravity effects disappeared at high heat and mass fluxes, however. The effects of convective motion and evaporation on the heat transfer were seen locally along the tube wall. For the nine gravity levels tested, transition regimes were observed at different flow and heating conditions. In microgravity, there was a noticeable reduction in heat transfer due to the weak turbulent mixing, both between bubbles and within the wake of bubbles. The average heat transfer coefficient along a 6 mm ID tube continuously decreased with decreasing gravity level for consistent flow regimes at low flow rate and low heat flux. The effect of gravity level on heat transfer decreased as both mass flux and heat flux increased. Due to the conflicting effect between buoyancy and inertial forces, churn flow and mixed regimes were observed in downward flow compared to microgravity and upward flow where the flow behavior was mostly bubbly flow. Using downward flow as a comparison to upward flow and microgravity flow is not recommended, but the pronounced effect of the flow instabilities during downward flow proved effective in studying local heat transfer phenomena using temperature sensitive paint. References Campbell, B. T., Liu, T., and Sullivan, J. P., 1994, “Temperature Sensitive Fluorescent Paint Systems,” AIAA Paper No. 94-2483.

	Ozawa, H., Laurence, S. J., Martinez Schramm, J., Wagner, A., and Hanne- mann, K., 2015, “Fast-Response Temperature-Sensitive-Paint Measurements on a Hypersonic Transition Cone,” Exp. Fluids, 56(2–13), p. 1853.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Abstracts for Journals and Proceedings	Al Hashimi H, Hammer C, Lebon MT, Kim J. "Phase Change Heat Transfer Measurements Using Optical Techniques." 9th World Conference on Experimental Heat Transfer, Fluid Mechanics, and Thermodyanamics (EXHFT), Iguazu Falls, Brazil, June 11-15, 2017. Conference Proceedings. 9th World Conference on Experimental Heat Transfer, Fluid Mechanics, and Thermodyanamics (EXHFT), Iguazu Falls, Brazil, June 11-15, 2017. , Jun-2017
Abstracts for Journals and Proceedings	Kim J, Lebon M, Hammer C. "Gravity effects on flow boiling heat transfer using temperature sensitive paints." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Al Hashimi H, Hammer C, Lebon MT, Zhang D, Kim J. "Phase change heat transfer measurements using temperature sensitive paints." Journal of Heat Transfer. 2018 Mar;140(3):031601-031601-15. https://doi.org/10.1115/1.4038135 , Mar-2018